

MIE1517 Sample Final Example

Instructor: Sinisa Colic

Date: N/A

Duration: 150 Minutes

Aids Allowed: None

Student Number: _____

Family Name(s): _____

Given Names(s): _____

*Do **not** turn this page until you have received the signal to start.
In the meantime, please read the instructions below carefully.*

This test consists of 9 questions on 19 pages (including this one), printed on both sides of the paper. *When you receive the signal to start, please make sure that your copy of the test is complete, fill in the identification section above, and write your name on the back of the last page.*

Answer each question directly on the test paper, in the space provided. If you need more space for one of your solutions, use the extra pages at the end of the test paper and *indicate clearly the part of your work that should be marked.*

Write up your solutions carefully! If you are giving only one part of an answer, indicate clearly what you are doing. Part marks might be given for incomplete solutions where it is clearly indicated what parts are missing.

MARKING GUIDE

1: ____/10

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TOTAL: ____/88

Question 1. [10 MARKS]

Circle the best answer for each of the questions below.

Part (a) [1 MARK]

Dropout is an example of ...

- (A) A way to prevent overfitting.
- (B) An optimizer.
- (C) A loss function.
- (D) A data augmentation technique.
- (E) A neural network architecture.

Part (b) [1 MARK]

Which of the following does not evaluate to 0?

- (A) sigmoid(0)
- (B) tanh(0)
- (C) relu(-5)
- (D) relu(0)
- (E) leaky_relu(0)

Part (c) [1 MARK]

Consider embedding a tweet by summing the GloVe embeddings of each word in a tweet. Which of the following is true about the embeddings of the tweets "i am happy" and "i am happy i am happy i am happy"?

- (A) The embeddings are equal.
- (B) The embeddings are close when measured using the Euclidean distance, but not the cosine similarity.
- (C) The embeddings are close when measured using the cosine similarity, but not the Euclidean distance.
- (D) The embeddings are close when measured using either the Euclidean distance or the cosine similarity.
- (E) None of the above.

Part (d) [1 MARK]

Which line of code performs the forward pass?

- (A) pred = model(xs)
- (B) loss = criterion(pred, ys)
- (C) loss.backward()
- (D) optimizer.step()
- (E) optimizer.zero_grad()

Part (e) [1 MARK]

How many floating-point multiplication operations are performed when applying a 3 x 3 convolution on a RGB image of size 6 x 6, with stride 1 and no padding?

- (A) 27
- (B) 48
- (C) 16
- (D) 9
- (E) None of the above

Part (f) [1 MARK]

Why do we need to pad sequences in an RNN?

- (A) We don't. Padding is for fully-convolutional networks.
- (B) We don't. Padding is for convolutional networks with some fully-connected layers.
- (C) To ensure all input sequences in a mini-batch have the same length.
- (D) To ensure all input sequences in the training set have the same length.
- (E) To ensure all input sequences in the validation set have the same length.

Part (g) [1 MARK]

Which part of a neuron has signals that do not attenuate with distance travelled?

- (A) soma
- (B) synapse
- (C) dendrite
- (D) axon
- (E) None of the above

Part (h) [1 MARK]

Which of the following is not an example of a distributed encoding of information?

- (A) The RNN embedding of a sentence.
- (B) The output of the encoder of an autoencoder.
- (C) The output of the generator of a GAN.
- (D) The GloVe embedding of a word.
- (E) All of the above are distributed encodings.

Part (i) [1 MARK]

Which of these is not a drawback of dimensionality reduction?

- (A) information loss
- (B) leads to poor visualization of data
- (C) it can be computationally expensive
- (D) adds complexity to the machine learning pipeline
- (E) transformed features are often difficult to interpret

Part (j) [1 MARK]

Which of the following about fairness of machine learning models is false?

- (A) A model can have high demographic parity, but poor individual fairness.
- (B) There is no agreed-upon definition of fairness.
- (C) Removing a sensitive feature from the data might not ensure disparate impact.
- (D) Optimizing for fairness can decrease model accuracy.
- (E) None of the above are false.

Question 2. [10 MARKS]

Circle either “True” or “False” for each of the below statements.

- a. ☒ True ☐ False A convolutional neural network typically has at least one fully-connected layer.
- b. ☐ True ☒ False A word2vec model is trained by mapping words into vector representations, and penalizing the distance between the representations of words that appear close to each other.
- c. ☒ True ☐ False The dimensionality of queries has to match the dimensionality of the keys?
- d. ☒ True ☐ False You can use mean-squared error loss for a classification tasks.
- e. ☒ True ☐ False The Multi-Head Attention is the most important layer in a Transformer.
- f. ☐ True ☒ False The generator and the discriminator of a GAN usually have the same number of parameters.
- g. ☐ True ☒ False RNNs are well suited for parallelization.
- h. ☒ True ☐ False Both CNN and RNN architectures use weight-sharing.
- i. ☐ True ☒ False Character-level RNN's are not capable of learning the long-term dependency to generate grammatically correct English sentences.
- j. ☒ True ☐ False Decreasing the temperature setting of an RNN generator increases sample quality.

Question 3. [10 MARKS]

Provided below are confusion matrices obtained from two different classification models: ANN Model and SVM Models. The models were trained to classify three types of flowers: setosa, versicolor and virginica based on a test dataset.

ANN Model - Confusion Matrix			
True Label	setosa	12	1
	versicolor	0	12
	virginica	0	1
		setosa	versicolor
		Predicted Label	

SVM Model - Confusion Matrix			
True Label	setosa	13	0
	versicolor	0	10
	virginica	0	0
		setosa	versicolor
		Predicted Label	

Part (a) [2 marks]

Which of the two models has the best accuracy?

1pt: some work related to the calculation of accuracy (~84%)

1pt: both models have the same accuracy

Part (b) [2 marks]

Why is accuracy not a good way to describe the performance of the models?

2pt: Key point, accuracy does not make a distinction on how well the model does on the different classes. If it does well on most classes then it could mask the poor performance on a few classes.

Part marks for accuracy does not work well when you have unbalanced data.

Part (c) [4 marks]

Describe how the performance differs between the two classifier models? Which of the two classifiers do you think has a better performance? Provide arguments for your choice?

1pt: SVM model makes one mistake of misclassifying versicolor as virginica

1pt: ANN model has varying degrees of misclassification for all flowers, but does better on classification of versicolor

1pt: both models have the same accuracy

1pt: Any justification for why ANN Model or SVM model is better

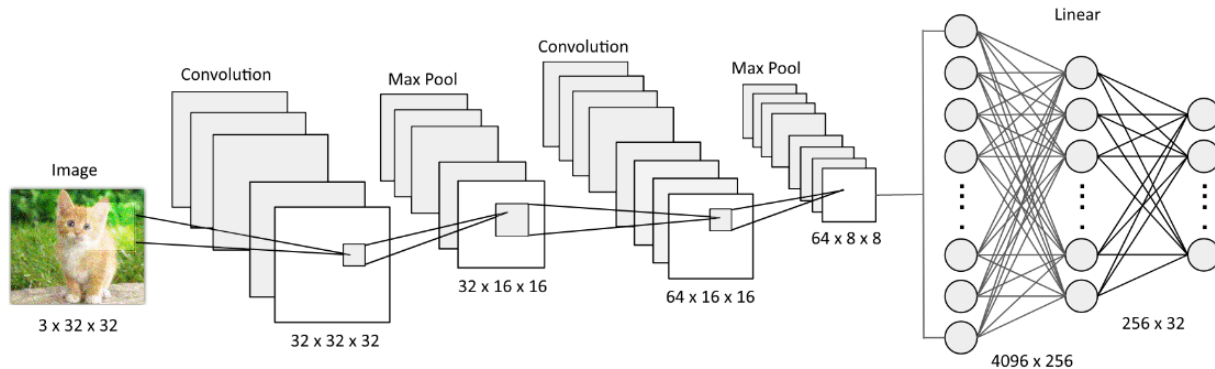
Part (d) [2 MARKS]

Why is it important to have a balanced dataset?

2pt: If the training data is imbalanced the model may learn to output the class that is more prevalent.

Question 4. [10 MARKS]**Part (a)** [6 marks]

Complete the PyTorch code to achieve the architecture described in the diagram below. You may assume that all the necessary packages/modules are already imported, and that the code below has no syntax errors.



```
class PetClassifier(nn.Module):
    def __init__(self):
        super(PetClassifier, self).__init__()
        #Conv2d: in_ch, out_ch, kernel_size, stride, padding
        self.conv1 = nn.Conv2d(_3_, _32_, _1_, _1_, _0_)      1pt
        #Pool: kernel_size, stride
        self.pool = nn.MaxPool2d(_2_, _2_) #kernel, stride  1pt
        self.conv2 = nn.Conv2d(_32_, _64_, _3_, _1_, _1_)    1pt
        self.fc1 = nn.Linear(_4096_, _256_) #input, output  1pt
        self.fc2 = nn.Linear(_256_, _32_)                    1pt

    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = x.view(_-1_, _4096_)                             1pt
        x = F.relu(self.fc1(x))
        x = self.fc2(x)
        return x
```

Part (b) [4 marks]

What changes would need to be made in the architecture and/or training code to allow this model to solve a regression problem?

2pt: The loss function would need to be changed to MSE

2pt: fc2 output would need to be changed from 32 to 1

Question 5. [10 MARKS]

Consider the following implementation of an autoencoder that is to colourize 32 x 32 greyscale images. You may assume that all the necessary packages/modules are already imported, and that the code below has no syntax errors.

```
class Autoencoder(nn.Module):
    def __init__(self):
        super(Autoencoder, self).__init__()

        self.encoder = nn.Sequential(
            nn.Conv2d(in_channels=1, out_channels=16, kernel_size=3, stride=2, padding=1),
            nn.ReLU(),
            nn.Conv2d(in_channels=16, out_channels=32, kernel_size=3, stride=2, padding=1),
            nn.ReLU(),
            nn.Conv2d(in_channels=32, out_channels=64, kernel_size=7, padding=0)
        )

        self.decoder = nn.Sequential(
            nn.ConvTranspose2d(in_channels=64, out_channels=32, kernel_size=7, padding=0),
            nn.ReLU(),
            nn.ConvTranspose2d(in_channels=32, out_channels=16, kernel_size=3,
                               stride=2, padding=1, output_padding=1),
            nn.ReLU(),
            nn.ConvTranspose2d(in_channels=16, out_channels=3, kernel_size=3,
                               stride=2, padding=1, output_padding=1),
            nn.Sigmoid()
        )
```

Part (a) [5 MARKS]

How many parameters are there in the above autoencoder? Show all of your work.

2.5pt: $(1 * 3 * 3) * 16 + 16$
 $(16 * 3 * 3) * 32 + 32$
 $(32 * 7 * 7) * 64 + 64$
 2.5pt: $(64 * 7 * 7) * 32 + 32$
 $(32 * 3 * 3) * 16 + 16$
 $(16 * 3 * 3) * 3 + 3$

Part (b) [2 MARKS]

What does the setting `output_padding=1` in the module `nn.ConvTranspose2d` do?

2pt: Adjust the output dimension by adding 1 to the height and width

Part (c) [3 MARKS]

The autoencoder tends to have blurry output images. Why does this happen and how can a GAN address this issue?

1.5pt: This results from using a MSE loss function that has an averaging effect on pixels

1.5pt: A GAN can correct this by having a neural network (i.e. discriminator) to replace the loss function and detect blurry images

Question 6. [10 MARKS]

We wish to train a Recurrent Neural Network to perform sentiment analysis. You may assume that all the necessary packages/modules are already imported, and that the code below has no syntax errors.

```
class SentimentAnalysis (nn.Module):
    def __init__(self, vocab_size, hidden_size, n_layers=1):
        super(SentimentAnalysis, self).__init__()
        self.ident = torch.eye(vocab_size)
        self.rnn = nn.GRU(input_size=vocab_size,
                           hidden_size=hidden_size,
                           num_layers=n_layers,
                           batch_first=True)
        self.decoder = nn.Linear(in_features=hidden_size,
                                   out_features=vocab_size)

    def forward(self, inp, hidden):
        inp = inp.view(1, -1)
        inp = self.ident[inp]
        output, hidden = self.rnn(inp, hidden)
        logits = self.decoder(output.squeeze(0))
        return logits, hidden
```

Part (a) [2 MARKS]

Assuming that the above code works as intended. What is the output of this piece of code?

```
>>> model = SentimentAnalysis (vocab_size=10, hidden_size=16)
>>> inp = torch.tensor([1, 0, 3, 2]) # a sequence with 4 tokens
>>> hidden = torch.randn(1, 1, 16)
>>> out, _ = model(inp, hidden)
>>> out.shape
```

2pt: `torch.Size([4, 10])`

Part (b) [2 MARKS]

What does each of the elements of the tensor **out** from part (a) represent?

2pt: $\text{out}[i,j]$ = unnormalized probability that the output at index i (the i -th token) should be a particular output token represented by vocabulary index j

Part (c) [2 MARKS]

What changes would need to be made to the code above when going from a GRU to a LSTM architecture?

1pt: need to initialize and add cell state (i.e. another hidden state)

1pt: change layer from `nn.GRU` to `nn.LSTM`

Part (d) [4 MARKS]

How could you use a RNN to achieve sentiment analysis on videos? What in the above sample could would need to change to facilitate sentiment analysis of videos?

2pt: Apply a CNN model to represent frames in a video

2pt: The code would be similar, only the input would be changed to represent different frames in the video instead of characters.

Question 7. [8 MARKS]

In this question, you will be working with the following Generative Adversarial Network (GAN):

```
D = Discriminator() #discriminator model
d_criterion = nn.BCEWithLogitsLoss()
```

```
G = Generator() #generator model
g_criterion = nn.MSELoss()
```

The following is a sample of training code for the discriminator and generator. The variable `images` contains a mini-batch of 32 training images. The generator takes a random vector of size 200 as input.

```
# images = mini batch data, with batch size 32
noise = torch.randn(32, 100)
inputs = torch.cat([G(noise), images])
outputs = D(inputs)
labels = torch.cat([torch.zeros(32),
                    torch.ones(32)])
d_loss = d_criterion(outputs, labels)
d_loss.backward()

noise = torch.randn(32, 100)
outputs = G(noise)
labels = torch.ones(32)
g_loss = g_criterion(outputs, labels)
g_loss.backward()
```

Part (a) [3 MARKS]

Identify three “unique” issues with the code above and how to correct them. Pseudocode is acceptable.

1. `g_criterion = nn.MSELoss()` -> `nn.BCEWithLogitsLoss()`
2. `random` change from 100 to 200
3. `outputs = G(noise)` -> `D(G(noise))`

Part (b) [3 MARKS]

We discussed that the balance between the discriminator and generator is important in training a GAN. Why is it a problem if the discriminator model is too accurate or too good? What are some things we can do to improve the GAN performance?

1pt: If the discriminator is too good then nothing the generator will do can fool the discriminator and hence it has nothing to learn from

2pt: Many options work: update the discriminator parameters less frequently than generator, anything that limits the performance of the discriminator to allow the generator to catch up

Part (c) [2 MARKS]

Do autoencoder models suffer from mode collapse? Why or why not? Explain.

1pt: No,

1pt: the autoencoder is used to reconstruct images, so it will try to reconstruct something similar to the class presented as input.

Question 8. [10 MARKS]**Part (a)** [4 MARKS]

How is a Transformer different from an RNN?

2pt: The main difference is that the Transformer takes in the entire sequence and can operate in parallel unlike RNNs that examine the input sequentially.

2pt: Positional encoder captures the sequential information, self-attention, etc.

Part (b) [3 MARKS]

How can the inclusion of attention mechanism help with debugging the model?

3pt: The alignment scores make the model easier to debug and interpret: for example, if the model makes a mistake, you can look at which part of the input it was paying attention to, and this can help diagnose the issue.

Part (c) [3 MARKS]

What are some problems with recurrent architectures?

3 pt: Not parallelizable across instances, difficulty with modelling long-term dependences, vanishing/exploding gradients. Note LSTM can help, but still these challenges persist as sequences get longer.

Question 9. [10 MARKS]

Consider the model you built in Lab Assignment 4 for spam detection. Suppose that we wish to build a spam blocking software to be used with our email?

Part (a) [3 MARKS]

What does it mean for a model to have “demographic parity”? List two demographic groups for which our model from Lab Assignment 4 likely does not achieve demographic parity.

1 pt: samples have equal representation of different demographics such as race, wealth, gender, etc.

2 pt: there could be many possible answers here, so just check to make sure it makes sense, for example, age of people typing emails, as people get older it is less likely that they write emails and would not be represented in the “ham” group, similarly those who may be poor would not be represented because they less likely to have access to computers and email, etc.

Part (b) [2 MARKS]

What are two things we can do to mitigate the issues in Part (a)?

+2 pt: Lots of options, anything that make sense is fine

- Gather more data from unrepresented groups
- Change the weighting of some of the data

Part (c) [5 MARKS]

In lecture we covered ethical case studies related to the use of AI systems with self-driving cars. In the case of the crash of Uber's self-driving car that resulted in death. What are some ethical concerns of this technology? Your responses should be in terms of the five ethical principles discussed in class. Indicate the most relevant ethical principles and provide a statements on how they apply to this case study.

+1 pt: responsibility – +1.5 not well defined who is responsible for the death of the pedestrian, Uber, the driver, or the programmer

+1 pt: transparency – +1.5 not clear how the AI is making the decision to run over the pedestrian

On this page, please write nothing except your name.

Family Name(s): _____

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